

Control Plan Case Studies - Real World Examples

Lessons Learned | Quality Head Agents

Case 1: CC Not Controlled - Customer Return

Problem: Weld tensile strength (CC, S=9) was in Control Plan with sample size "1 per shift" and gauge "visual inspection". A batch with sub-standard welds passed and reached the customer - vehicle recall risk. Root Cause: CC characteristic allowed on sample-only visual inspection. Detection method was not capable of detecting sub-surface weld defects. Lesson: CC characteristics must have 100% detection OR a validated poka-yoke (automated sensor). Sample inspection is not acceptable for CC. Auditor raised Major NC - Clause 8.5.1.1.

Case 2: Reaction Plan Not Followed

Problem: SPC chart on bore diameter showed 3 consecutive points near upper control limit (Nelson Rule 2). Operator did not react - continued running. 200 parts produced, 45 were out of spec. Root Cause: Reaction plan said "inform supervisor" - no specific action defined. Supervisor was unavailable. Operator continued. Lesson: Reaction plan must not rely on verbal notification only. Must specify: Stop machine, check last X pieces, measure, record, tag suspect material. Written process, not verbal.

Case 3: CP Not Updated After Change

Problem: Engineering changed bore diameter tolerance from +/-0.05 to +/-0.02. Control Plan was not updated - still showed old tolerance. QA was inspecting to old spec for 3 weeks. Customer found deviation during audit. IATF Minor NC raised. Lesson: Engineering changes must trigger mandatory CP review before implementation. Implement a change control process: Engineering Change Note (ECN) must include QA sign-off confirming CP updated.

Case 4: CP Linked to Wrong PFMEA

Problem: During IATF audit, auditor found Control Plan referenced PFMEA Rev 2, but current PFMEA was Rev 5 (updated after 2 customer complaints). The CP did not reflect the strengthened controls in PFMEA Rev 5. Major NC - Clause 8.5.1.1. Lesson: When PFMEA is updated, CP must be reviewed and updated simultaneously. Both documents must carry the same revision level for linked process steps. Document control system must enforce this linkage.